

PUTREE: A PHOTOREALISTIC LARGE-SCALE VIRTUAL BENCHMARK FOR FOREST TRAINING

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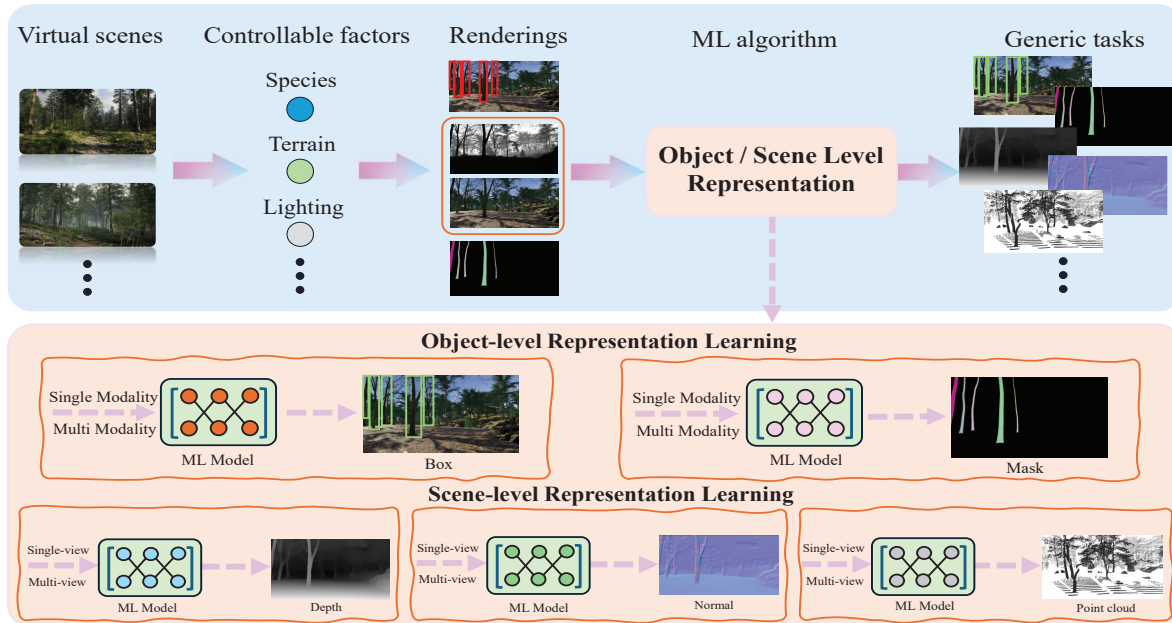


Figure 1: **Overview of our newly proposed *PUTree* dataset.** (a) Overall pipeline diagram showing how the application creates controllable virtual 3D forest scenes and renders various frames for ML algorithm training. (b) Our dataset can be utilized for both object-level and scene-level representation learning. The algorithms can take either single or multi-modality (RGB and depth) and single or multiple viewpoints (image or video frames) as input. The dataset and simulation tool can also be used to enable other forest applications such as automatic DBH measurement, 3D forest mapping and reconstruction, etc.

ABSTRACT

Forest systems play an important role in mitigating anthropogenic climate change and regulating global climate. However, due to difficulties in collecting wild data and lack of forestry expertise, the availability of large-scale forest datasets is very limited. In this work, we establish a new virtual forest dataset named *PUTree*. Our goal is to create a larger, more photo-realistic and diverse dataset as a powerful training resource in the wild forest. Early experimental results demonstrate its validity as a new forest benchmark for the evaluation of tree detection and segmentation algorithms, and its potential to serve as virtual training environments in VR/AR applications.

Index Terms: Computing methodologies—Modeling and simulation—Simulation support systems—Simulation environments; Computing methodologies—Artificial intelligence—Computer vision—Image segmentation / Object detection.

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1 INTRODUCTION

Tree detection, segmentation, and reconstruction in wild scenes are fundamental to accurate forest mapping and monitoring, and has gained much attention [2–4, 6, 7, 9]. However, due to the expense and difficulty of collecting a sufficient amount of annotated data within wild forests, current forest datasets [5, 8] are rather limited in number and include inevitable labelling errors inherent in human annotation. Therefore, we propose that a good forest dataset should avoid human annotation and create as many photorealistic annotations as possible. The dataset could provide realistic 3D models and rendered images of trees, terrain, etc. that could be imported into VR/AR applications to create immersive simulated natural environments for education, gaming, measurement, etc. The high level of detail in the models could make the virtual forests seem very realistic.

In this work, we present a new forest simulation tool and *PUTree* dataset for training algorithms in different forest environments (Fig. 1). To better verify the usage value of the proposed dataset, we compare the results of training detection and segmentation methods with and without our data. The evaluation shows that our data improves the accuracy of the recent model [1]. Our project page and sample data will be available at <https://forestdataset.github.io/PUTREE/>.

Using our simulation tool and data, a person without a strong forest background is now able to generate a large number of different labeled synthetic images, including the information of tree count,

species, masks, scene depth, and point cloud. We have described the creation of our dataset in Sec. 2, the extensibility and potential use scenarios in Sec. 3, and a conclusion with future work in Sec. 4. For its efficacy, we hope our work can catalyze a shift in simulation systems and digital twins based on synthetic data in digital forestry.

2 PUTREE DATASET

Manual annotation in wild scenes typically requires a lot of time (3-5 minutes per image) and effort, especially for densely populated forests. Moreover, it introduces some unavoidable mistakes in tree boundary and occlusion handling. We constructed our simulated virtual forest dataset to seamlessly and efficiently train a variety of machine learning algorithms for forest tasks. The dataset includes a set of high-quality photo-realistic tree models for 19 tree species, customizable terrain, weather, and lighting using Unreal Engine.

2.1 Dataset Creation

To precisely segment the trunk regions of interest from the collected high-quality tree models, we use Autodesk Maya to separate each tree model in FBX or OBJ format into trunks, branches, and leaves, and only render the trunks for the segmentation camera. With individual tree models of different species, we populate the trees in the designated terrain map with randomly assigned trunk density and location. Within the same terrain, we define a set of camera trajectories and move the multi-sensor camera along the trajectory to obtain a set of collections (e. g. , RGB, depth, mask) under the same timestamp. In this way, it ensures the correct alignment of multi-modal image collections. As a post-processing stage to generate the COCO annotation files, we adopt OpenCV to automatically detect object contours and bounding boxes and export them as our final outputs. As a result, we are able to obtain 50k video frames.

2.2 Metrics and Evaluations

Standard metrics are used to measure the usability of the generated dataset, including the mean average precision (mAP) at the IoU threshold of 50%. From the preliminary results in Figure 2, it can be observed that the real-world segmentation and detection performance is significantly improved when using our dataset for pre-training, especially for tree trunk masks, revealing opportunities for improvement in unstructured forests and other forest-like scenarios.

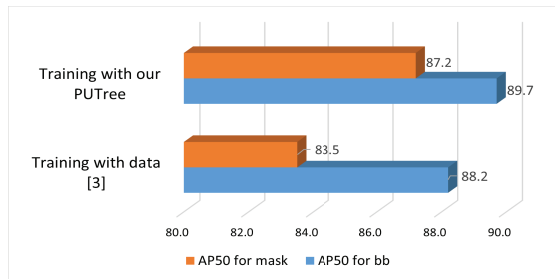


Figure 2: Segmentation and detection evaluation on 100 real images collected in Canadian forests [3]. The algorithms are greatly improved by incorporating the dataset we created in pre-training.

3 USAGE SCENARIOS

The *PUTree* dataset enables us to train data-hungry algorithms for several forest-understanding tasks, avoid over-fitting to small-size data, and evaluate them using highly credible ground truths.

Automatic Forest DBH Measurement: Measuring tree diameter at breast height (DBH) is a primary task in forestry. Given RGB

and/or depth, we use the previously mentioned model trained on the *PUTree* for tree segmentation, and then estimate tree DBH using segmentation masks and depth maps. While traditional methods such as tape measurements often introduce errors and inconsistencies into the DBH ground truth, our data provide accurate measurement ground truth derived from tree models. This allows for a more convincing evaluation of the model’s predictions, thereby increasing the reliability and validity of the DBH estimate.

Forest 3D Mapping and Reconstruction: Forest mapping and reconstruction is critical for accessing information about tree types, locations, and health. Forestry inventory methods, especially in large-scale and GNSS-disturbed forest environments, cannot guarantee the collection of high-quality and reliable data. The *PUTree* dataset, on the other hand, provides a solution by recording accurate camera parameters across multiple trajectories and point clouds from diverse forest environments, which enhances algorithm development and evaluation. In addition, the controllable and varied forest environment settings enhance algorithmic capabilities in understanding and interpreting complex real-world forest scenarios.

4 CONCLUSION

In this work, we proposed a new *PUTree*, a high-quality and large-scale virtual forest dataset with diverse, controllable and photo-realistic rendering data from the powerful Unreal Engine. Our dataset can be used in a wide range of forestry applications, including tree counting, detection, segmentation, localization, and 3D mapping. We hope that these efforts could facilitate more intelligent and digital advances in the field of forestry. For future work, we plan to conduct user studies with forest experts on the fidelity and practical values, and develop bundled applications based on the generated datasets.

5 ACKNOWLEDGMENTS

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